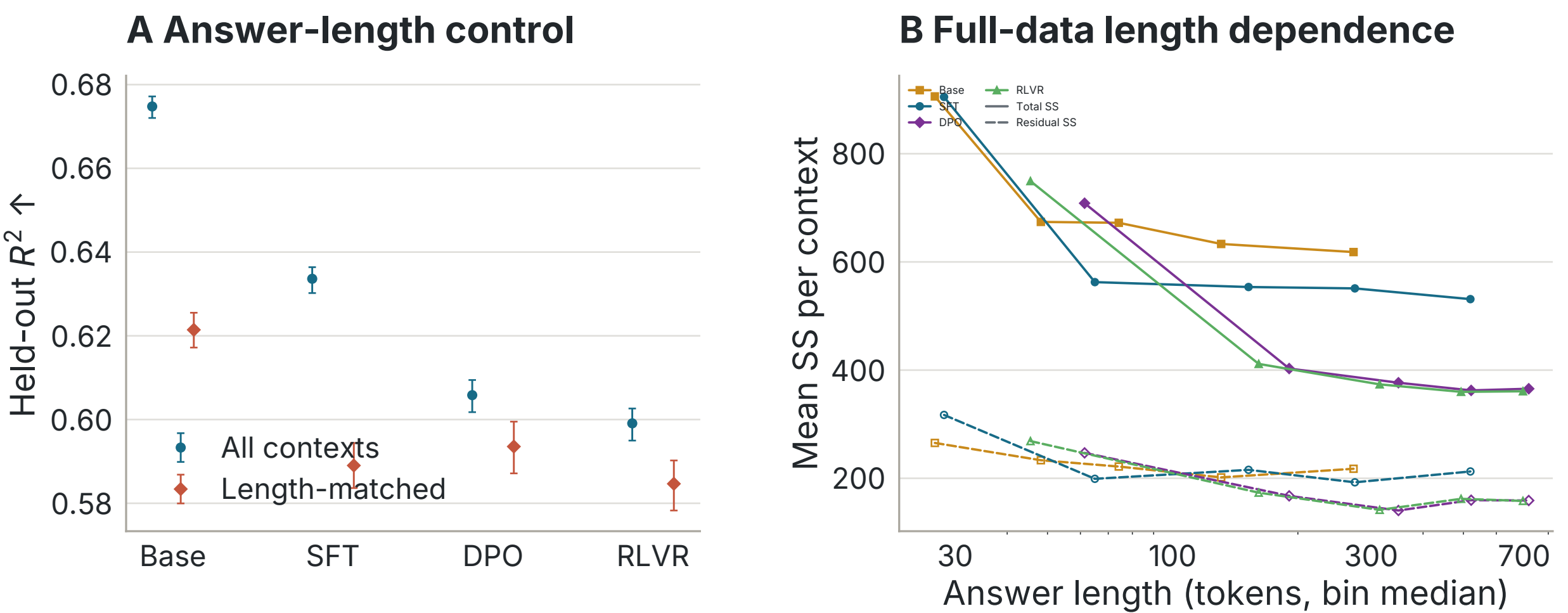

A Answer-length control
Held-out R^2 ↑
0.68
0.66
0.64
0.62
0.60
0.58
All contexts
Length-matched
Base
SFT
DPO
RLVR
B Full-data length dependence
Mean SS per context
800
600
400
200
Base
SFT
DPO
RLVR
Total SS
Residual SS
30
100
300
700
Answer length (tokens, bin median)